

Cisco Packet Tracer

VoIP Network Setup

Step by Step Configuration Tutorial

CallManager Express | DHCP Option 150 | Voice VLAN | IP Phones

Goal: Build a small company VoIP network in Packet Tracer where two Cisco 7960 IP phones register to Router0 using CallManager Express and call each other using extensions 1001 and 1002.

Overview

This guide walks through a complete Voice over IP setup in Cisco Packet Tracer. The network uses Router0 as the DHCP server and CallManager Express server. Switch0 connects the two IP phones and supports the voice VLAN. At the end of the lab, IP Phone1 should show extension 1001, IP Phone2 should show extension 1002, and the phones should be able to place calls to each other.

What You Need

Quantity	Device	Model	Purpose
1	Router0	Cisco 2811	DHCP server and CallManager Express server
1	Switch0	Cisco 2960 24TT	Layer 2 switch for phone connections and voice VLAN
2	IP Phone1 and IP Phone2	Cisco 7960	VoIP endpoints for extension 1001 and 1002

Network Topology and Addressing

Device	Model	IP Address	Role
Router0	2811	192.168.10.1	CME, DHCP, TFTP address through DHCP Option 150
Switch0	2960 24TT	N/A	Layer 2 switch and voice VLAN support
IP Phone1	7960	DHCP from 192.168.10.0/24	Extension 1001
IP Phone2	7960	DHCP from 192.168.10.0/24	Extension 1002

Port Connections

Use these exact ports for the Packet Tracer build.

Connection	From	To	Cable Type
Router uplink	Router0 FastEthernet0/0	Switch0 FastEthernet0/24	Copper Straight Through
Phone 1	IP Phone1 FastEthernet0	Switch0 FastEthernet0/1	Copper Straight Through
Phone 2	IP Phone2 FastEthernet0	Switch0 FastEthernet0/2	Copper Straight Through

Important: This corrected version uses Switch0 FastEthernet0/1 for IP Phone1, FastEthernet0/2 for IP Phone2, and FastEthernet0/24 for the Router0 uplink.

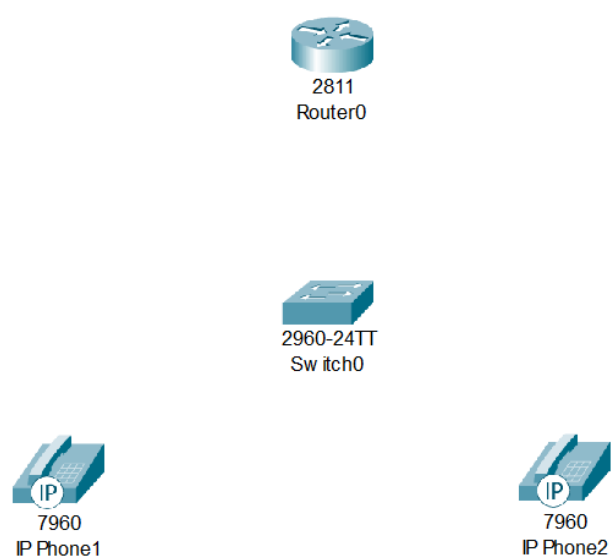


Figure 1. Starting topology with Router0, Switch0, IP Phone1, and IP Phone2.

1. Build the Topology

Open Cisco Packet Tracer and start a new blank project by selecting File > New. Place the devices on the canvas in a simple top down layout with the router above the switch and the two phones below the switch.

1. Go to Network Devices > Routers and drag a Cisco 2811 router onto the canvas. The device name should be Router0.
2. Go to Network Devices > Switches and drag a Cisco 2960 24TT switch below the router. The device name should be Switch0.
3. Go to End Devices > IP Phone and drag two Cisco 7960 IP Phones onto the canvas. The device names should be IP Phone1 and IP Phone2.

Layout Tip: Keep the topology organized so Router0 is on top, Switch0 is in the middle, and the phones are on the left and right. This makes the lab easier to read and easier to troubleshoot.

2. Connect the Devices

Use Copper Straight Through cables for all connections. Select the lightning bolt icon at the bottom of Packet Tracer, then choose the Copper Straight Through cable.

4. Connect Router0 FastEthernet0/0 to Switch0 FastEthernet0/24.
5. Connect IP Phone1 FastEthernet0 to Switch0 FastEthernet0/1.
6. Connect IP Phone2 FastEthernet0 to Switch0 FastEthernet0/2.

Note: The links may show red or orange at first. This is normal before the router interface, DHCP, telephony service, VLAN, and phone power are fully configured.

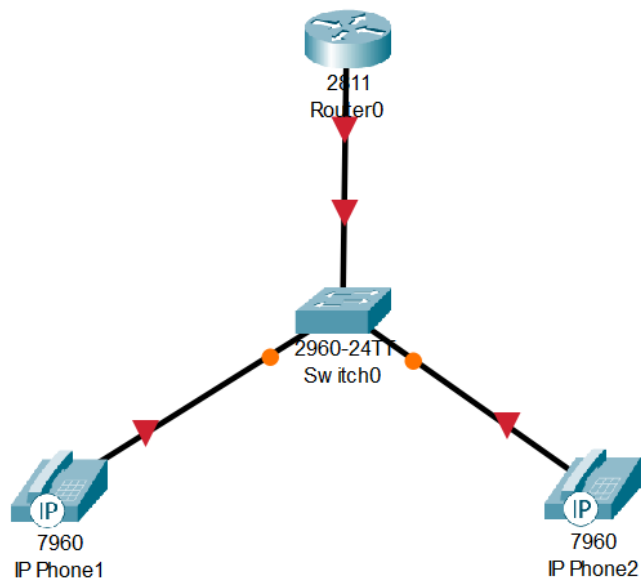


Figure 2. Devices connected using Router0 Fa0/0, Switch0 Fa0/24, Switch0 Fa0/1, and Switch0 Fa0/2.

3. Configure DHCP on Router0

Router0 must provide IP addresses to the phones and tell the phones where to find the TFTP and CME server. DHCP Option 150 points the phones to Router0 at 192.168.10.1.

Before You Start: Click Router0, open the CLI tab, and press Enter. If Packet Tracer asks whether to enter the initial configuration dialog, type no and press Enter.

```
enable
configure terminal
hostname Router0

interface FastEthernet0/0
  ip address 192.168.10.1 255.255.255.0
  no shutdown
exit

ip dhcp excluded-address 192.168.10.1

ip dhcp pool VOIP
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  option 150 ip 192.168.10.1
exit
```

Checkpoint: After the no shutdown command, you should see messages showing FastEthernet0/0 changed state to up. That means the router interface is active.

4. Configure CallManager Express on Router0

CallManager Express is the phone service that allows the Cisco IP phones to register and call each other. The ip source-address must match the Router0 FastEthernet0/0 address.

```
telephony-service
  max-ephones 5
  max-dn 5
  ip source-address 192.168.10.1 port 2000
  auto assign 1 to 5
  create cnf-files
exit

ephone-dn 1
  number 1001
exit

ephone-dn 2
  number 1002
exit
```

Packet Tracer Note: If you see a message saying the clock is not set or synchronized when you run create cnf-files, the lab can still work. The phones can still register.

5. Configure Voice VLAN on Switch0

Switch0 needs a voice VLAN on the ports connected to the IP phones. The router uplink port must also be placed in VLAN 10 so Router0 can communicate with the phone network.

```
enable
configure terminal
hostname Switch0

vlan 10
  name VOICE
exit

interface FastEthernet0/1
  switchport mode access
  switchport voice vlan 10
exit

interface FastEthernet0/2
  switchport mode access
  switchport voice vlan 10
exit

interface FastEthernet0/24
  switchport mode access
  switchport access vlan 10
exit
```

Port Summary: FastEthernet0/1 is for IP Phone1, FastEthernet0/2 is for IP Phone2, and FastEthernet0/24 is for Router0.

```
%LINK-S-CHANGED: Interface FastEthernet0/2, changed state to up
%LINK-S-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Switch0
Switch0(config)#vlan 10
Switch0(config-vlan)#name VOICE
Switch0(config-vlan)#exit
Switch0(config)#interface FastEthernet0/1
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport voice vlan 10
Switch0(config-if)#exit
Switch0(config)#interface FastEthernet0/2
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport voice vlan 10
Switch0(config-if)#exit
Switch0(config)#interface FastEthernet0/24
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport access vlan 10
Switch0(config-if)#exit
Switch0(config)#
```

Figure 3. Switch0 configured with VLAN 10, phone ports, and the Router0 uplink port.

6. Power On the IP Phones

Cisco 7960 IP phones in Packet Tracer need a power adapter before they can boot. Do this for both phones.

7. Click IP Phone1 and open the Physical tab.
8. Drag the power adapter into the power socket on the back of the phone.
9. Repeat the same process for IP Phone2.
10. Wait about 30 to 60 seconds while the phones boot, request DHCP, and register with Router0.

Expected Result: After registration, IP Phone1 should show extension 1001 and IP Phone2 should show extension 1002 in the GUI tab.

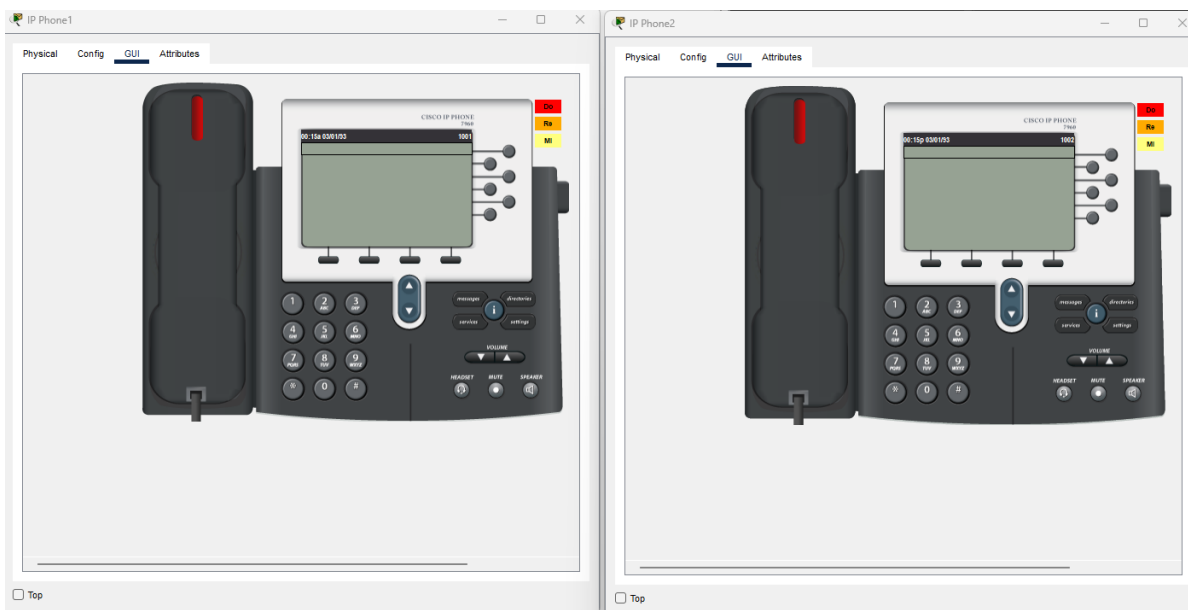


Figure 4. Both Cisco 7960 phones are registered. IP Phone1 shows 1001 and IP Phone2 shows 1002.

7. Make a Test Call

After both phones are registered, test the VoIP setup by calling from IP Phone1 to IP Phone2.

11. Click IP Phone1 and open the GUI tab. Confirm the phone screen shows 1001.
12. Pick up the handset or click the speaker button to get a dial tone.
13. Dial 1002 using the phone keypad.
14. IP Phone2 should ring.
15. Open IP Phone2 in the GUI tab and answer the call.
16. Both phones should show a connected call.

Success Criteria: The lab is successful when IP Phone1 can call IP Phone2 and IP Phone2 can answer the call.

8. Save the Configuration

Save the running configuration on both Router0 and Switch0 so the work stays saved inside the Packet Tracer file.

Router0 save command

```
enable
copy running-config startup-config
```

Switch0 save command

```
enable
copy running-config startup-config
```

Final Save: Also save the Packet Tracer project by selecting File > Save As. A clear file name is Small_Company_VoIP_Setup.pkt.

Troubleshooting Guide

Problem	What to Check
Phone screen is blank	Open the phone Physical tab and make sure the power adapter is installed.
Phone says configuring IP	Check that Router0 DHCP is configured and that Router0 Fa0/0 is up.
Phone says unregistered	Check DHCP Option 150, telephony-service, ip source-address, and create cnf-files.
Phone does not show extension	Check ephone-dn 1 and ephone-dn 2. Make sure auto assign 1 to 5 is configured.
No dial tone	Wait 30 to 60 seconds, then verify both phones are registered.
Call does not connect	Make sure IP Phone1 has extension 1001 and IP Phone2 has extension 1002.
Links stay orange or red	Check cabling and verify Switch0 Fa0/1, Fa0/2, and Fa0/24 were configured correctly.

Final Verification Checklist

- Router0 FastEthernet0/0 is configured as 192.168.10.1/24 and is up.
- Router0 DHCP pool VOIP uses network 192.168.10.0/24.
- DHCP Option 150 points to 192.168.10.1.
- CallManager Express is enabled with max-ephones, max-dn, and ip source-address.
- Extension 1001 is assigned to IP Phone1.
- Extension 1002 is assigned to IP Phone2.
- Switch0 VLAN 10 is created and named VOICE.
- Switch0 Fa0/1 and Fa0/2 use switchport voice vlan 10.
- Switch0 Fa0/24 is placed in VLAN 10 for the Router0 uplink.
- Both phones are powered on and registered.
- IP Phone1 can call IP Phone2 by dialing 1002.

Appendix: Full Copy and Paste Commands

Router0 Full Configuration

```
enable
configure terminal
hostname Router0

interface FastEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

ip dhcp excluded-address 192.168.10.1

ip dhcp pool VOIP
```



```
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
option 150 ip 192.168.10.1
exit

telephony-service
max-ephones 5
max-dn 5
ip source-address 192.168.10.1 port 2000
auto assign 1 to 5
create cnf-files
exit

ephone-dn 1
number 1001
exit

ephone-dn 2
number 1002
exit

end
copy running-config startup-config
```

Switch0 Full Configuration

```
enable
configure terminal
hostname Switch0

vlan 10
name VOICE
exit

interface FastEthernet0/1
switchport mode access
switchport voice vlan 10
exit

interface FastEthernet0/2
switchport mode access
switchport voice vlan 10
exit

interface FastEthernet0/24
switchport mode access
switchport access vlan 10
exit

end
copy running-config startup-config
```